

Pedersen Vector Commitments and Inner Product Arguments

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1 Background

The classical Pedersen commitment scheme enables committing to a scalar a with perfect hiding and computational binding. Given a random scalar r , the commitment is defined as:

$$C = aG + rH$$

where G and H are independent generators of a prime order group. The hiding property is unconditional, while binding relies on the hardness of the discrete logarithm problem.

2 Pedersen Vector Commitments

Extending Pedersen commitments to vectors, for scalars $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$, we define the commitment:

$$P = \sum_{i=1}^n (a_i G_i + b_i H_i) + rG$$

where $\mathbf{G} = (G_1, \dots, G_n)$ and $\mathbf{H} = (H_1, \dots, H_n)$ are vectors of independent generators. The inclusion of a random r maintains perfect hiding even in the multi-dimensional setting.

3 Inner Product Argument

Given a commitment P to vectors $\mathbf{a}$ and $\mathbf{b}$, the goal is to prove knowledge of these vectors such that:

$$\langle \mathbf{a}, \mathbf{b} \rangle = c$$

for some known scalar c , without revealing $\mathbf{a}$ or $\mathbf{b}$. Here, the inner product is defined as:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i.$$

4 -Protocol for the Inner Product Argument

The proof proceeds as a -protocol, using recursive vector compression:

- **Split:** The prover divides the vectors into halves:

$$\mathbf{a} = (\mathbf{a}_L, \mathbf{a}_R), \quad \mathbf{b} = (\mathbf{b}_L, \mathbf{b}_R).$$

- **Commit:** Two auxiliary commitments are generated:

$$L = \sum_{i=1}^{n/2} (a_{L,i} G_{R,i} + b_{R,i} H_{L,i}) + r_L G,$$

$$R = \sum_{i=1}^{n/2} (a_{R,i} G_{L,i} + b_{L,i} H_{R,i}) + r_R G.$$

- **Challenge:** The verifier sends a random challenge $x \in \mathbb{Z}_q^*$.
- **Compress:** The prover computes new compressed vectors:

$$\mathbf{a}' = x\mathbf{a}_L + x^{-1}\mathbf{a}_R, \quad \mathbf{b}' = x^{-1}\mathbf{b}_L + x\mathbf{b}_R,$$

and updated generators:

$$G'_i = G_{L,i}^x G_{R,i}^{x^{-1}}, \quad H'_i = H_{L,i}^{x^{-1}} H_{R,i}^x.$$

The randomness is also updated:

$$r' = r_L x + r_R x^{-1}.$$

- **Repeat:** The process recurses until the vectors are reduced to a single element.

5 Final Commitment and Verification

At the final step, the prover presents a commitment:

$$P' = aG + bH + rG$$

for scalars a , b , and r . The verifier checks that the recomputed commitment matches P adjusted by all L , R , and challenge contributions, ensuring correctness of the claimed inner product.

6 Security Properties

6.1 Perfect Hiding

The commitment scheme remains perfectly hiding, even throughout recursive compression, due to the continuous injection of randomness and the verifier's unpredictable challenges.

6.2 Computational Binding

Binding holds under the discrete logarithm assumption. Forging two different openings to the same commitment would imply solving a discrete log problem.

6.3 Logarithmic Proof Size

Each round halves the vector size, yielding a proof size that grows logarithmically with the dimension n . This ensures efficiency even for large vectors.

6.4 Soundness

The challenge-response structure guarantees that a cheating prover succeeds only with negligible probability. If the prover does not know valid openings satisfying the inner product constraint, it will almost certainly fail verification.

7 Conclusion

We have described a Pedersen-based vector commitment scheme that supports efficient proofs of inner products between committed vectors. Using a recursive-protocol structure, the scheme achieves perfect hiding, computational binding, and logarithmic proof size. These properties make it highly suitable for applications requiring compact zero-knowledge proofs of linear relations over committed data.